Songwei Ge

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Research Interest

Image and Video Synthesis, AI for Creativity, Out-of-Domain Generalization and Adversarial Robustness

Education

Department of Computer Science, University of Maryland *College Park, MD, USA*

Ph.D. in Computer Science *Sep. 2020 - Present*

- GPA: 4.00/4.00
- Selected courses: Foundation of Deep Learning, Advanced Numerical Optimization, Advanced Techniques in Visual Learning and Recognition, Fundamental Concepts of Differential Geometry.
- Advisors: David Jacobs and Jia-Bin Huang.

School of Computer Science, Carnegie Mellon University *Pittsburgh, PA, USA*

M.S. in Computational Biology *Aug. 2018 - May. 2020*

- GPA: 4.15/4.30
- Selected courses: Machine Learning for Ph.D., Convex Optimization, Probabilistic Graphical Models.
- Thesis Advisors: Barnabás Póczos, Eunsu Kang and Ziv Bar-Joseph.

School of Information, Renmin University of China *Beijing, China*

B.S. in Computer Science and Technology *Sep. 2014 – Jul. 2018*

- Thesis Advisors: Zhicheng Dou. (**Best thesis award** in Computer Science.)

HillHouse Academy, Renmin University of China *Beijing, China*

B.S. in Finance *Sep. 2014 – Jul. 2018*

- Thesis Advisor: Wei Xu.

Research Experience

Computer Science Department, University of Maryland, College Park *College Park, MD, USA*

Advisor: David Jacobs *Sep. 2020 – Present*

- Studied out-of-domain generalization of self-supervised models and proposed a hard-negative mining strategy to improve robustness.
- Theoretically and empirically showed that shift invariance can lead to weaker adversarial robustness.

Machine Learning Department, Carnegie Mellon University *Pittsburgh, PA, USA*

Advisors: Eunsu Kang and Barnabás Póczos *Oct. 2018 – May. 2020*

- Worked on generative models for 3D objects, especially for point clouds and meshes when specialized data is limited.

Machine Learning Department, Carnegie Mellon University *Pittsburgh, PA, USA*

Advisors: Ziv Bar-Joseph, Zachary Lipton and Barnabás Póczos *Jan. 2019 – May. 2020*

- Studied the robustness of CNN-based models under statistical and adversarial distribution shift and proposed a regularization technique on the local predictive power to improve the model generalization under out-of-domain settings.

Big Data Analytics and Intelligence Lab, Renmin University of China *Beijing, China*

Advisors: Zhicheng Dou and Ji-Rong Wen *Jan. 2017 – Feb. 2018*

- Proposed a hierarchical LSTM model that models long-term (entire history) and short-term (present session) user profiles and significantly improves search results quality with personalization on a large-scale query log.

Work Experience

NVIDIA Research *Santa Clara, CA, USA*

Advisors: Yogesh Balaji, Ming-Yu Liu *May. 2022 -*

- Work on large-scale video synthesis.

Facebook AI Research *Menlo Park, CA, USA*

Advisor: Devi Parikh *June. 2021 – Dec. 2021*

- Developed a position-agnostic VQVAE and a position-dependent transformer for long and high-resolution video generation.

Facebook AI Research *Menlo Park, CA, USA*

Advisors: Vedanuj Goswami, Larry Zitnick, Devi Parikh *June. 2020 – Aug. 2020*

- Collected two datasets of creative sketches; each contains 10k sketches along with part annotations. Proposed DoodlerGAN – a part-based Generative Adversarial Network (GAN) – to generate unseen compositions of novel part appearances. Developed a web demo for the project.

Microsoft Research Xiaoice Science Team *Beijing, China*

Advisor: Ruihua Song *Apr. 2018 – Jul. 2018*

- Designed contextualized frameworks to add sound effects to a radio story, which achieved smaller false positives than classic methods.

Selected Publications

In Preparation

- **Hyperbolic Contrastive Learning for Visual Representations beyond Objects.**
  Songwei Ge*, Shlok Mishra*, Simon Kornblith, Chun-Liang Li, David Jacobs.
  ***In Submission to NeurIPS**, 2022*

Peer Reviewed Publications

- **Long Video Generation with Time-Agnostic VQGAN and Time-Sensitive Transformer.**
  Songwei Ge, Thomas Hayes, Harry Yang, Xi Yin, Guan Pang, David Jacobs, Jia-Bin Huang, Devi Parikh.
  *In European Conference on Computer Vision (**ECCV**), 2022.*
- **MUGEN: A Playground for Video-Audio-Text Multimodal Understanding and GENeration.**
  Thomas Hayes, Songyang Zhang, Xi Yin, Guan Pang, Sasha Sheng, Harry Yang, Songwei Ge, Isabelle Hu, Devi Parikh
  *In European Conference on Computer Vision (**ECCV**), 2022.*
- **Robust Contrastive Learning Using Negative Samples with Diminished Semantics.**
  Songwei Ge, Shlok Mishra, Haohan Wang, Chun-Liang Li, David Jacobs.
  *In Neural Information Processing Systems (**NeurIPS**), 2021.*
- **Shift Invariance Can Reduce Adversarial Robustness.**
  Songwei Ge*, Vasu Singla*, Ronen Basri, David Jacobs.
  *In Neural Information Processing Systems (**NeurIPS**), 2021.*
- **Conceptual Blending with Large-scale Language and Vision Models.**
  Songwei Ge, Devi Parikh.
  *In International Conference on Computational Creativity (**ICCC Oral**), 2021.*
- **Creative Sketch Generation.**
  Songwei Ge, Vedanuj Goswami, Larry Zitnick, Devi Parikh.
  *In Workshop on Machine Learning for Creativity and Design at NeurIPS (**Oral**), 2020*
  *In International Conference on Learning Representations (**ICLR**) 2021.*
- **Supervised Adversarial Alignment of Single-Cell RNA-seq Data.**
  Songwei Ge, Haohan Wang, Amir Alavi, Eric P. Xing, Ziv Bar-Joseph.
  *In the 24th International Conference on Research in Computational Molecular Biology (**RECOMB**), 2020*
- **Learning Robust Global Representations by Penalizing Local Predictive Power.**
  Haohan Wang, Songwei Ge, Eric P. Xing, Zachary C. Lipton.
  *In Neural Information Processing Systems (**NeurIPS**), 2019.*
- **From Text to Sound: a Preliminary Study on Retrieving Sound Effects to Radio Stories.**
  Songwei Ge, Curtis Xuan, Ruihua Song, Chao Zou, Wei Liu and Jin Zhou.
  *In ACM International Conference on Research and Development in Information Retrieval (**SIGIR**), 2019.*
- **Hallucinating Point Cloud into 3D Sculptural Object.**
  Songwei Ge*, Chun-Liang Li*, Eunsu Kang*, Lingyao Zhang, Austin Dill, Manzil Zaheer, Barnabás Póczos.
  *In Workshop on Machine Learning for Creativity and Design at NeurIPS, 2018, and Inter-Society for the Electronic Arts (**ISEA**), 2019*
- **A Computational Strategy for Finding Novel Targets and Therapeutic Compounds for Opioid Dependence.**

Xiaojun Wu, Siwei Xie, Lirong Wang, Peihao Fan, Songwei Ge, Xiang-Qun Xie, Wei Wu.
In PLoS One, 2018

- **Personalizing Search Results Using Hierarchical RNN with Query-aware Attention.**
  Songwei Ge, Zhicheng Dou, Zhengbao Jiang, Jian-Yun Nie, Ji-Rong Wen.
  *In ACM International Conference on Information and Knowledge Management (**CIKM**), 2018*

Selected Honors & Awards

2019-2020	**CMU MSCB Academic Scholarship**	*Pittsburgh, PA*
Jun 2018	**Outstanding Undergraduate Thesis Award** (Top 1% in RUC & Top 1 in the Computer Science)	*Beijing, China*
Jan 2017	**Meritorious Winners,** American Mathematical Contest in Mathematical Modeling	*Beijing, China*
2014-2018	**RUC School of Information Scholarship**	*Beijing, China*

Professional Service

2022	**External Reviewer** The ACM Symposium on User Interface Software and Technology	*UIST*
2022	**Conference Reviewer** European Conference on Computer Vision	*ECCV*
2022	**Emergency Reviewer** Conference on Computer Vision and Pattern Recognition	*CVPR*
2022	**Conference Reviewer** International Conference on Machine Learning	*ICML*
2022-2023	**Conference Reviewer** International Conference on Learning Representations	*ICLR*
2021-2022	**Conference Reviewer** Annual Conference on Neural Information Processing Systems	*NeurIPS*

Teaching

2020	**Teaching Assistant** Introduction to Machine Learning	*College Park, MD*
2019	**Guest Lecturer** Art and Machine Learning	*Pittsburgh, PA*
2017	**Teaching Assistant** Advanced Algorithm Design	*Beijing China*

Skills

Programming and Deep Learning Libraries	Python, C, C++, C#, R, MATLAB, Java, Javascript, HTML, Tensorflow, PyTorch, Jax
Others	Qualified basketball referee